Edge-Efficient WiFi Indoor Localization via Lightweight Deep Learning and Knowledge Distillation

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***Abstract*—Accurate WiFi indoor localization is increasingly required for edge AI applications such as autonomous robots, smart factories, healthcare monitoring, and intelligent buildings. However, state-of-the-art deep learning models remain computationally expensive, limiting their deployment on resource-constrained devices. This paper investigates whether lightweight deep neural networks, combined with knowledge distillation, can achieve competitive localization performance while substantially reducing computational cost. Using the public WILD dataset, we systematically evaluate four lightweight student architectures—MobileNetV2, MobileNetV2-UNet, TinyCNN, and Mamba—against the Deep localization teacher (DLoc) model in terms of localization accuracy, robustness to corrupted Channel State Information (CSI), cross-session generalization, statistical stability, and computational efficiency. Extensive experiments revealed that the best-performing model, a distilled MobileNetV2-UNet, contains only 2.41 million parameters, providing a 6.8× reduction in model size and requiring 56× fewer FLOPs than DLoc while matching its published median localization error of 0.64 m. Furthermore, we reveal that the benefits of knowledge distillation are capacity-dependent, improving robustness and cross-session generalization rather than in-distribution localization accuracy, with significant gains observed only for student models containing approximately two million parameters or more. The proposed model consistently achieves the best zero-shot performance on unseen environments while remaining suitable for deployment on resource-constrained edge devices. These findings demonstrate that highly accurate, robust, and computationally efficient WiFi indoor localization can be achieved without large deep neural networks, providing practical guidelines for the design of edge-ready localization systems.**

***Index Terms*—Indoor localization, WiFi, Channel State Information (CSI), knowledge distillation, CNN, U-Net.**

I. Introduction

GPS has long provided reliable positioning outdoors, but its accuracy vanishes inside buildings. Indoor localization has therefore been an active research topic for the last two decades, supporting applications that range from tracking factory assets [1] to guiding robots and monitoring patients.

The leading approach is WiFi-based localization, which analyzes Channel State Information (CSI) from access points [2] to obtain precise estimates of the time-of-flight (ToF) and angle-of-arrival (AoA) of incoming signals—sufficient for sub-meter accuracy. The state of the art, DLoc [2], uses a deep learning model to translate this CSI data into a heatmap of the device's location on the floor plan.

Deep learning has achieved strong results in dense prediction and intelligent sensing through effective feature representations and carefully designed optimization objectives, including Gaussian-weighted feature modeling for monocular depth estimation, appearance-aware optimization for dense prediction, and nonlinear loss integration for novel-view synthesis [3]–[5]. Building on these architecture- and objective-driven design principles, the present work extends them to CSI-based indoor localization, targeting lightweight architectures and knowledge distillation for resource-constrained edge AI systems.

However, DLoc has a significant limitation: it requires a large training dataset and a ResNet model with 16.5 million parameters—too demanding for standard gateways, access points, or smartphones. We investigate whether "student" models—smaller, more efficient architectures—can deliver comparable performance without this overhead.

The main contributions of this work are summarized as follows:

- We design and evaluate lightweight deep neural network architectures for CSI-based indoor localization, enabling deployment on resource-constrained edge devices while significantly reducing computational complexity.
- We investigate the effectiveness of knowledge distillation across student models with different capacities and identify a capacity threshold below which distillation no longer provides meaningful performance improvements.
- We perform a comprehensive evaluation of localization accuracy, robustness to corrupted CSI, cross-session generalization, statistical stability, and computational efficiency using the public WILD benchmark dataset.
- We demonstrate that a distilled MobileNetV2-UNet achieves localization accuracy comparable to DLoc while reducing model size by 6.8×, computational complexity by 56×, and CPU inference time by approximately one order of magnitude, making it well suited for edge AI deployment.

II. Related Work

Indoor localization has been extensively studied using WiFi Channel State Information (CSI), owing to its ability to exploit existing wireless infrastructure without requiring dedicated positioning hardware. Among deep learning-based approaches, DLoc [2] demonstrated that CSI measurements can be effectively transformed into localization heatmaps using convolutional neural networks, achieving sub-meter localization

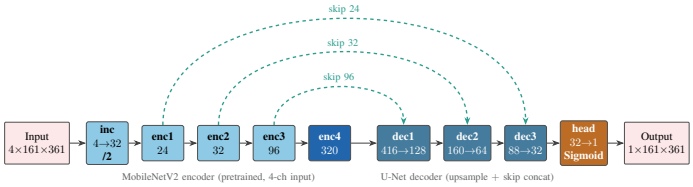


Fig. 1. The MobileNetV2-UNet student. A MobileNetV2 encoder (light blue) downsamples the four-channel input; a decoder (dark blue) upsamples back to full resolution; teal dashed arcs are U-Net skip connections bypassing the bottleneck.

accuracy. More recent studies have explored transformer-based architectures for non-line-of-sight environments [6], meta-learning [7], few-shot learning for long-term environmental drift [8], domain adaptation [9], adaptive angle-of-arrival estimation [10], explainable localization [11], and data augmentation techniques [12]. Although these methods have significantly improved localization accuracy and adaptability, they generally rely on increasingly complex deep neural networks, making deployment on resource-constrained edge devices challenging.

To address this challenge, recent research has focused on lightweight deep learning and edge AI. Efficient architectures such as MobileNetV2 [13], residual learning [14], and encoder–decoder networks with skip connections [15] have demonstrated excellent accuracy-to-complexity trade-offs across a wide range of vision tasks. More recently, state-space models such as Mamba [16], [17] have emerged as computationally efficient alternatives to transformer architectures. In parallel, lightweight inference has become a key requirement for green wireless sensing [18] and TinyML-enabled IoT systems [19], highlighting the growing importance of efficient deep learning models for real-world deployment.

Knowledge distillation has become one of the most effective approaches for compressing deep neural networks by transferring knowledge from a high-capacity teacher model to a lightweight student model [20]. Beyond image classification, knowledge distillation has been successfully applied to dense prediction tasks such as object detection [21] and more recently to cross-modal localization [22]. However, existing studies primarily focus on improving prediction accuracy through knowledge transfer. The interaction between student model capacity, knowledge distillation, robustness to corrupted CSI, and cross-session generalization for WiFi indoor localization remains largely unexplored.

Unlike previous work, this paper systematically investigates edge-efficient WiFi indoor localization by jointly studying lightweight neural architectures and knowledge distillation. Rather than proposing another localization backbone, we analyze the trade-offs among localization accuracy, computational complexity, robustness, and cross-session generalization across multiple lightweight student models. This provides practical design guidelines for deploying deep learning-based indoor localization on resource-constrained edge devices.

III. Methodology

A. Overview of the DLoc Framework

DLoc [2] formulates CSI-based localization as a heatmap prediction problem. Four AoA–ToF images, each corresponding to one access point, are processed by a ResNet encoder followed by two decoder branches: a location decoder that predicts the localization heatmap and a consistency decoder that estimates an offset map. The complete network contains approximately 16.5 million parameters.

B. WILD Dataset

The WILD dataset, collected in Jacobs Hall at the University of California, San Diego (UCSD), comprises five data collection sessions. The two July sessions provide 17,574 training samples and 4,260 in-distribution test samples. Three additional August sessions introduce progressively more challenging environmental changes, including rearranged furniture (11,201 samples), different furniture configurations (8,754 samples), and the addition of an aluminum reflector (5,486 samples).

To evaluate cross-session generalization, we adopt the leave-one-session-out (LOSO) protocol proposed in the original DLoc work [2]. Specifically, the dataset is divided into three folds, each using the two July sessions together with two August sessions for training, while the remaining August session is reserved exclusively for testing.

C. Student Network Architectures

To investigate the trade-off between localization accuracy and computational efficiency, four lightweight student architectures were considered. All student models use the same input and output as the teacher model. Table I lists the students that achieved the best results on the dataset, along with the computation required to run them. MobileNetV2 (2.27 M parameters) utilizes depthwise separable convolutions; MobileNetV2-UNet (2.41 M) has the same architecture with the addition of skip connections that directly concatenate feature maps between the encoder and decoder networks (as seen in Fig. 1). For comparison, we additionally evaluate two highly compact architectures: Mamba, a 0.65 M-parameter state-space model, and TinyCNN, an extremely lightweight 47 K-parameter encoder–decoder network. These models enable us to investigate the lower end of the model-capacity spectrum and identify the point at which aggressive model compression begins to compromise localization performance. Their experimental results are discussed in Section V-A.

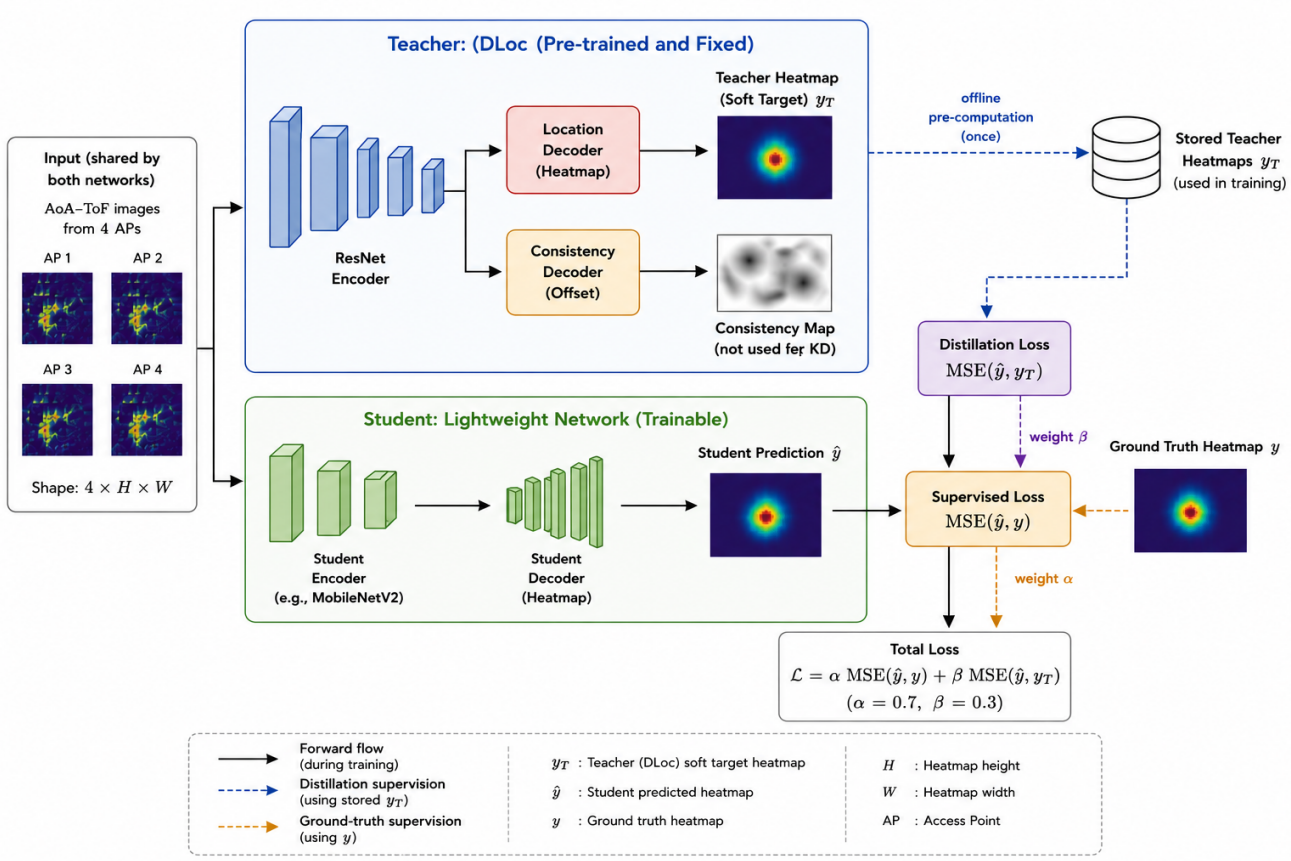


Fig. 2. Overall framework of the proposed edge-efficient WiFi indoor localization system. Four AoA–ToF images from the WILD dataset are simultaneously processed by a fixed DLoc teacher and a lightweight student network. The teacher generates a soft localization heatmap that is pre-computed offline and used as the distillation target. During training, the student is optimized using a weighted combination of supervised loss with the ground-truth heatmap and knowledge distillation loss with the teacher heatmap. During inference, only the lightweight student network is deployed, enabling efficient edge localization with substantially reduced computational complexity.

TABLE I
HIGH-PERFORMING MODELS AND THEIR COMPUTATIONAL COST (BATCH SIZE 1; DLOC = TEST-TIME PATH: ENCODER + LOCATION DECODER).

Model	Params	Compr.	FLOPs	CPU (ms)	GPU (ms)
DLoc (teacher)	16.5M	1×	81.06G	270.9	16.2
MobileNetV2	2.27M	7.2×	1.28G	17.9	6.3
MobileNetV2-UNet	2.41M	6.8×	1.45G	27.2	4.5

D. Knowledge-Distillation Framework

As illustrated in Fig. 2, the proposed framework follows a teacher–student paradigm for efficient CSI-based indoor localization. A pre-trained DLoc model serves as the teacher, while a lightweight neural network acts as the student. During training, both networks receive the same four-channel AoA–ToF input. The teacher generates soft localization heatmaps that are pre-computed offline and used as additional supervision, whereas the student learns from both the ground-truth heatmaps and the teacher's predictions. During inference, only the lightweight student network is retained, enabling efficient deployment on resource-constrained edge devices.

For a given input, the fixed DLoc teacher produces a soft target heatmap y_T, each student trains both independently (supervised only by the ground-truth label y) and in a distilled mode minimizing

$$\mathcal{L} = \alpha\,\mathrm{MSE}(\hat{y}, y) + \beta\,\mathrm{MSE}(\hat{y}, y_T), \tag{1}$$

where the coefficients α and β are empirically set to 0.7 and 0.3. The teacher outputs are pre-computed offline, so the teacher itself never runs during training of the students. The soft heatmaps produced by the teacher provide richer spatial supervision than hard ground-truth labels by encoding the confidence distribution over nearby candidate locations.

E. Evaluation Metrics

The localization error (e) is the Euclidean distance between the two heatmap peaks:

$$e = 0.1 \cdot \|\arg\max(\hat{y}) - \arg\max(y)\|_2 \ \ [\mathrm{m}], \tag{2}$$

where the factor of 0.1 converts pixels to meters. Following [2], the median of e is reported along with the 90th and 99th percentile values. Because the error is discrete and quantized to multiples of $0.1\sqrt{n}$, models with identical median localization errors may exhibit substantially different error distributions. Reporting higher-order percentiles therefore provides a more comprehensive characterization of localization performance.

IV. Implementation

All models are implemented in PyTorch. TinyCNN and MobileNetV2 are trained with the Adam optimizer with a learning rate of 10^{-4} and batch size of 64 for 50 epochs; MobileNetV2-UNet is trained for 120 epochs using cosine-annealing and step-decay schedulers; and DLoc employs the Adam optimizer with β_1=0.5 and a StepLR scheduler with γ=0.9. The best models from each architecture are selected based on the lowest test error. All training is performed on rented NVIDIA H100 GPUs, though the latency measurements in Table I are collected on a single desktop RTX 3060 GPU. Due to the teacher not being retrained within the leave-one-session-out (LOSO) procedure, MobileNetV2-UNet runs in standalone mode during that evaluation to maintain comparability with reported DLoc results. All code, weights, training logs and pre-computed teacher outputs are archived in a public repository so every result can be independently reproduced.

V. Results and Discussion

The experiments are organized around four research questions on the design of edge-efficient WiFi indoor localization, evaluating localization accuracy, robustness to corrupted CSI, cross-session generalization, and the accuracy–efficiency trade-off.

A. In-Distribution Accuracy

This experiment evaluates whether lightweight students preserve DLoc's localization accuracy despite far fewer parameters and lower computational complexity.

Research Question (RQ1): Can lightweight deep neural networks achieve DLoc-level localization accuracy while significantly reducing model size and computational cost?

Table II reports the localization errors of the high-performing models on the held-out Session-2 data split; Fig. 3 plots the corresponding error CDFs. MobileNetV2 achieves the same result as our DLoc reproduction (median error of 0.707 m), yet uses 7.2× fewer parameters; this suggests that the original DLoc architecture is substantially over-parameterized for in-distribution localization. TinyCNN was not included since it could not achieve sufficiently low errors (3.92 m standalone, 3.71 m distilled), indicating a capacity floor of approximately 2 million parameters for this task. MobileNetV2-UNet achieves the lowest median error of 0.640 m, an error 9.5% lower than MobileNetV2 and our DLoc reproduction, and level with DLoc's published 0.64 m [2]. The standalone and distilled models exhibit the same median error: in-distribution, the teacher's outputs match the ground truth, so there is no benefit to distillation. This changes outside of distribution (Secs. V-B and V-C). Finally, Mamba was not competitive either (median error 10.56 m): its training and test errors rise together from the first epoch—not overfitting but an architectural mismatch, since flattening the 2D heatmaps into 1D sequences destroys spatial locality.

Among the highly compressed models, Mamba—despite roughly 14× more parameters than TinyCNN (0.65 M vs. 47 K)—produced the largest localization error, indicating that architectural suitability is at least as important as model capacity for CSI-based heatmap prediction.

TABLE II
In-Distribution Accuracy (Session-2 Test Set, N=4,260; Best in **Bold**; Published DLoc from [2]). TinyCNN and Mamba Excluded as Non-Competitive (See Text).

Model	Params	Mode	Med. (m)	P90 (m)	P99 (m)
DLoc (published)	16.5M	—	**0.64**	1.6	3.2
DLoc (our repro)	16.5M	—	0.707	∼1.6	∼4.0
MobileNetV2	2.27M	Standalone	0.707	1.616	3.992
MobileNetV2	2.27M	Distilled	0.707	1.612	4.003
MobileNetV2-UNet	2.41M	St. (Cos)	**0.640**	1.523	2.750
MobileNetV2-UNet	2.41M	St. (Step)	**0.640**	1.552	2.918
MobileNetV2-UNet	2.41M	Dist. (Cos)	**0.640**	**1.513**	**2.587**

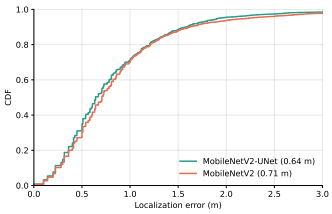


Fig. 3. CDF of in-distribution localization error for the high-performing models.

B. Robustness to Corrupted Inputs

Real deployments face imperfect wireless conditions, so we evaluate the robustness of the proposed models under CSI corruption scenarios never observed during training.

Research Question (RQ2): Which lightweight architecture provides the greatest robustness to corrupted CSI, and what role does knowledge distillation play in improving robustness?

We corrupt the CSI only at test time: per-AP and random AP dropout, additive Gaussian noise ($\sigma \in \{0.1, 0.2\}$), attenuation, and spatial blur. Table III shows the Gaussian-noise case.

Distillation provides robustness to high-capacity student models; the standalone MobileNetV2 degrades to 16.4 m with $\sigma = 0.1$, yet its distilled version degrades only to 4.8 m, which is 3.4× better. Skip connections provide an additional source of robustness against corruption; the standalone MobileNetV2-UNet achieves 3.0 m, which is 5.5× better than the standalone MobileNetV2. The two benefits are somewhat complementary; the distilled MobileNetV2-UNet does not benefit from the distillation process in relation to its robustness to Gaussian noise (3.000 m vs. 3.469 m), but it does benefit from the distillation in relation to its robustness to AP dropout. Distillation also appears to have a capacity threshold; on the 47 K TinyCNN it leaves the robustness to noise unchanged (9.46 m vs. 9.44 m), yet degrades its robustness to AP dropout (random 1-AP: 7.9 m to 13.6 m): the small model spends its capacity imitating how the teacher combines the APs rather than on robustness to their dropout at test time.

C. Cross-Session Generalization

Practical systems must generalize to environments unseen during training; we evaluate this under environmental change using the leave-one-session-out protocol.

TABLE III
GAUSSIAN-NOISE ROBUSTNESS OF THE HIGH-PERFORMING MODELS ($\sigma = 0.1$)

Model	Mode	Median (m)	P90 (m)
MobileNetV2	Standalone	16.368	20.939
MobileNetV2	Distilled	4.769	12.310
MobileNetV2-UNet	Standalone	**3.000**	**8.310**
MobileNetV2-UNet	Distilled	3.469	10.473

TABLE IV
LEAVE-ONE-SESSION-OUT VS. PUBLISHED DLOC (MEAN±STD, 3 SEEDS; OURS = MOBILENETV2-UNET).

Held-out session	Med. (ours)	Med. ([2])	P90 (ours)	P90 ([2])
Aug16_1 (furn.)	0.852 ± 0.044	0.71	1.839 ± 0.067	1.71
Aug16_3 (diff. furn.)	0.822 ± 0.040	0.82	2.793 ± 0.140	2.52
Aug16_4_ref (refl.)	1.116 ± 0.016	1.05	2.489 ± 0.030	2.77
Mean	0.930	0.86	2.374	2.33

Research Question (RQ3): Can lightweight student networks generalize across unseen environments while maintaining localization performance comparable to DLoc? The LOSO results over three random seeds (Table IV) show that the 2.41 M-parameter MobileNetV2-UNet is competitive with the 16.5 M-parameter DLoc: it matches DLoc on the different-furniture fold (0.822 m vs. 0.82 m), is within 0.14 m of DLoc on the remaining folds, has essentially the same mean P90 (2.374 m vs. 2.33 m), and has a lower P90 on the challenging reflector fold (2.489 m vs. 2.77 m). The residual gap on the reflector fold is somewhat to be expected: DLoc's offset decoder regularizes estimates across APs—beneficial under specular reflection—and the MobileNetV2-UNet student has no such regularizer.

Beyond the LOSO comparison, we additionally train the models on only the two July sessions and evaluate zero-shot on the never-seen August sessions (Table V). The distilled U-Net has the lowest median and P90 on each of the August sessions; on the most difficult (reflector) session its median is 1.281 m, versus 1.334 m for the standalone model and 1.442 m for the best MobileNetV2. Thus, the distilled MobileNetV2-UNet is recommended as the deployment configuration.

D. Statistical Stability and Efficiency

Practical deployment also requires computational efficiency and stable performance across training runs, which we analyze next.

Research Question (RQ4): Do the proposed lightweight models provide a favorable accuracy–efficiency trade-off while maintaining statistically stable localization performance? To confirm the gains are not seed noise, the DLoc baseline was trained with three different seeds for the full 120 epochs. The median error of 0.875 ± 0.036 m (coefficient of variation of 4.1%) is actually higher than the value obtained after only 50 epochs, because the StepLR schedule (γ=0.9) never decays the learning rate to zero over the 120 epochs. Nevertheless, the MobileNetV2-UNet model's 0.640 m median is 6.5 standard deviations below this mean, indicating significance. The P99 value varies far more across the baseline seeds (coefficient of variation of 52%).

TABLE V
ZERO-SHOT CROSS-SESSION ACCURACY (JULY-ONLY TRAINING). CELLS: MEDIAN / P90 (M); BEST IN **BOLD**.

Model	Mode	Aug16_1	Aug16_3	Aug16_4_ref
MobileNetV2	Standalone	1.118 / 3.138	1.204 / 3.471	1.500 / 3.367
MobileNetV2	Distilled	1.118 / 3.228	1.204 / 3.342	1.442 / 3.711
MobileNetV2-UNet	Standalone	1.063 / 2.508	1.005 / 3.704	1.334 / 3.301
MobileNetV2-UNet	Distilled	**1.020 / 2.319**	**1.005 / 2.608**	**1.281 / 3.002**

Regarding efficiency (Table I), the MobileNetV2-UNet model uses far fewer GFLOPs (1.45 compared to DLoc's 81.06, or $56\times$ less), requires less time on CPU (27.2 ms compared to DLoc's 270.9 ms, or $10\times$ faster) and is $3.6\times$ faster on GPU. Thus, the gains in accuracy come at no computational cost, making the model a strong candidate for edge deployment.

VI. CONCLUSION

This paper investigated whether high localization accuracy in WiFi indoor positioning necessarily requires large deep neural networks. Through a systematic evaluation of lightweight student architectures, we showed that DLoc can be compressed by nearly an order of magnitude while maintaining localization performance comparable to the original model. In particular, MobileNetV2 (2.27 M parameters) matches the localization accuracy of our reproduced DLoc model, while MobileNetV2-UNet (2.41 M parameters) further achieves the published DLoc median localization error of 0.640 m using $6.8\times$ fewer parameters, $56\times$ fewer FLOPs, and approximately $10\times$ lower CPU inference time.

Beyond computational efficiency, our experiments reveal two important design insights. First, the effectiveness of knowledge distillation is capacity-dependent: it primarily improves robustness to corrupted inputs and cross-session generalization rather than in-distribution localization accuracy, with meaningful gains emerging only for student models containing approximately two million parameters or more. Second, architectural design plays a crucial role in localization robustness, as U-Net skip connections consistently improve resilience to noisy CSI while knowledge distillation is particularly beneficial under access-point dropout.

Overall, the distilled MobileNetV2-UNet provides the best balance between localization accuracy, robustness, generalization, and computational efficiency, making it the recommended architecture for practical edge deployment of WiFi indoor localization systems.

Future work will focus on validating the proposed approach on resource-constrained edge devices, collecting a CSI dataset in a different building to evaluate cross-building generalization, investigating corruption-aware training strategies, and extending the multi-seed statistical analysis to all student architectures.

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